

Goibibo: Reducing time to publish hotel and user- generated photo reviews with Google Cloud

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ABOUT GOIBIBO



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About Goibibo

Goibibo is one of India's leading online travel brands. The service enables travelers to book their hotel stay and travel on flights, trains, buses, and cars. Goibibo prides itself on providing the widest travel choices possible while delivering a fast, trusted user experience across search, booking, payment, settlement, and refunds.

Industries: Travel & Hospitality

photos on the Google Cloud platform from a few hours to 30 minutes and an average time for publish user-generated photo reviews from hours to 10 minutes.

Location: India

Google Cloud (<https://cloud.google.com/>)

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Cloud Vision API (<https://cloud.google.com/vision/>)

Cloud AutoML (<https://cloud.google.com/automl/>)

Google Cloud Results

- Supports forecasted growth to classify, moderate, and annotate 20,000 to 25,000 images on average per day within one year
- Reduces the average time for

Enables cost-effective scaling to process 12K+ images on average per day

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India's domestic and international travel market is growing rapidly – and Goibibo is poised to reap the rewards. Goibibo is a leading application in the travel category – enabling travelers to book hotels and travel on flights, trains, buses, and cars.

ML opportunities

The business is growing Goibibo in its primary market in India, South East Asian countries, and other countries where outbound tourism from India

is high. A recent report

(<https://www.bain.com/insights/how-does-india-travel/>)

prepared by Bain & Company revealed Indian travelers went on approximately 2 billion domestic and international trips in 2018 and spent nearly \$94 billion on transport, lodging, and consumption during their travels. The report also found that as internet penetration increases, more travelers will book online – rising to 35 percent by 2021.

As Goibibo looked to the future, it examined the opportunities that machine learning and AI present to improve the customer experience provided by its online and mobile properties. In particular, the business wanted to improve the moderation, classification, scene recognition, and annotation of photos provided by users when reviewing hotels on Goibibo and of photos provided by the hotels themselves for their profiles on the platform.

The existing system used for these purposes was time-consuming and manpower-intensive. Both users and hotels had to wait a few hours for their photos to be published.

"Google Cloud AI
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easy and
accurate to use.

Cloud AutoML
API in particular
yielded very
accurate
classification
results."

—Dhruv Pathak, Vice
President, Engineering,
Goibibo

Google Cloud products accurate

The business opted to supplement machine learning on Goibibo's content processing pipelines through two Google Cloud (<https://cloud.google.com/>) products – Cloud Vision API (<https://cloud.google.com/vision/>) and Cloud AutoML (<https://cloud.google.com/automl/>).

Cloud Vision API provides pre-trained machine learning models through APIs that enable the rapid assignment of labels to images and quick classification into predefined categories. The service allows businesses to detect objects and faces, read text, and build metadata into image catalogues. Cloud AutoML allows developers to train and deploy machine learning models swiftly.

"Google Cloud AI solutions are easy and accurate to use. Cloud AutoML API in particular yielded very accurate classification results," says Pathak.

Cloud AutoML and Cloud Vision API key

Goibibo uses Cloud AutoML to create and train a machine learning model to identify and categorize images such as photos of swimming pools, restaurants, or washrooms. "We are training the model using past data to appropriately classify and annotate all images received from hoteliers and users in future," says Pathak.

"Processing all these images through Google Cloud, Cloud Vision API, and Cloud AutoML provides two points of value to our business. First, when hotel owners want to add photos to their hotel profile on our platform, they can do so in 30 minutes on

average, rather than the few hours required previously. Second, we can process and send user reviews with hotel photos live more quickly and manifest that content more intelligently, which helps increase trust and bookings for positively-reviewed hotels."

*—Dhruv Pathak, Vice
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Goibibo*

The business uses Cloud Vision API to moderate, annotate, and classify images into suitable categories, and also to flag images that may be considered profane, incorporate nudity, or may otherwise be offensive/inappropriate for the

platform. Scene recognition information is also used to manifest image content in a more intelligent way on the platform.

Goibibo now processes about 12,000 images supplied by hotels and reviewers per day, with the number of images captured for Goibibo reviews now up to 4 million.

Adding photos in 30 minutes

"Processing all these images through Google Cloud, Cloud Vision API, and Cloud AutoML provides two points of value to our business," says Pathak. "First, when hotel owners want to add photos to their hotel profile on our platform, they can do so in 30 minutes on average, rather than the few hours required previously. Second, we can process and send user reviews with hotel photos live more quickly and manifest that content more intelligently, which helps increase trust and bookings for positively-reviewed hotels."

The business is publishing user hotel reviews with photos in 10 minutes on average, rather than the few hours required previously.

Saves time and manpower

Intelligent automation through Cloud Vision and Cloud AutoML has also enabled Goibibo to save time and manpower moderating, annotating, and categorizing images. Goibibo expects to increase the number of images processed per day to 20,000 to 25,000 over the next year or so, through a

combination of organic growth and the proposed launch of new features.

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